

# Recording a Route Without Satellites

Speed from Vibration and Direction from Turns on a Smartphone, Measured Against GPS

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## Abstract

When GPS is lost — in a tunnel, an underground car park or an aircraft — a phone can still draw the route it travels. This paper describes a system that reads *speed* from how the vehicle shakes and *direction* from the phone’s gyroscope, accelerometer and magnetometer, using GPS for nothing but the first point, and measures it against GPS on a motorcycle, in a car, on foot and on a plane. Every recording was replayed through the newest version from its raw sensor data, with GPS used only as the answer key: 74 motorcycle and car journeys (20 hours, 340 km that GPS could check), 45 straight stretches and seven longer walks on foot, and one flight. Direction was within  $30^\circ$  of GPS 73% of the time on a motorcycle, 83% in a car, 93% on foot and 99% in the air, and the median route was turned  $9^\circ$  on a motorcycle and  $16^\circ$  in a car. Distance came out 10% short on a motorcycle, within 1% in a car and 4% long on foot; in the air, measuring the takeoff counted 61% of the flight. A phone with no examples of its own starts with a small built-in network, which on recordings it had never seen was as accurate as a full store. We also report what each part of the direction method adds, where the system fails, and the approaches that did not work.

**Keywords:** dead reckoning; smartphone sensors; GNSS-denied navigation; vehicle speed estimation; heading estimation; magnetometer

## 1 Introduction

A phone that loses GPS still knows a great deal about how it moves. Its accelerometer feels every push and every bump, its gyroscope measures every turn, its magnetometer senses north, and its barometer feels height. In principle that is enough to keep drawing a route through a tunnel, an underground car park or a flight. In practice two things make it hard.

The first is *speed*. Adding up acceleration, the textbook answer, does not work on a phone: its small sensor errors grow into kilometres within minutes [1, 2]. The second is *direction*. The gyroscope measures turns but not north; the magnetometer measures north but in a vehicle it is disturbed by steel, and in any case it gives the direction the *phone* points, not the direction of travel, which differ by however the phone happens to sit in a pocket or a holder [3].

This paper describes a system built around those two problems, running as a mode of an ordinary iPhone app, and measures how well it works. Its contributions are:

- **A complete route recorder that uses GPS for the first point only.** Speed comes from a fingerprint of the vehicle’s vibration, looked up in a store of the phone’s own earlier trips or, on a new phone, answered by a small built-in network. Direction comes from the vehicle’s turns, which reveal how the phone sits, on top of a heading whose slow drift is measured at stops and whose starting error is corrected by the magnetometer whenever the magnetic field can be trusted (Section 3).

- **An evaluation against GPS that reflects the newest version.** Every recording was replayed through the current method from its raw sensor data, and every result is split by what was moving: a motorcycle, a car, a person walking or a plane (Sections 4 and 5). Appendix A lists every journey.
- **An account of what matters.** We measure what each part of the direction method adds (Section 5.5), where the system goes wrong, and the approaches that did not work (Section 6).

## 2 Related work

**Inertial navigation.** Integrating acceleration twice gives position, but the error grows quickly — with the square of time for an accelerometer bias and faster still for a gyroscope bias [1]; smartphone accelerometers and gyroscopes carry biases and noise large enough to make unaided integration useless within minutes [2]. Practical systems therefore constrain the problem.

**Pedestrian dead reckoning.** For walking, the usual constraint is the step: count steps, estimate their length and their direction [4]. A sensor on the foot can reset its velocity at every stance [5]; a phone in a pocket cannot, and its heading depends on how it is held, which several methods try to resolve [3]. Our walking mode follows the step-counting tradition; its direction method is our own.

**Orientation.** Orientation filters fuse gyroscope, accelerometer and magnetometer readings [6], and on an iPhone Apple’s Core Motion provides one. Such filters assume gravity is the only steady force, so a vehicle’s sustained acceleration — a takeoff roll — is mistaken for a tilt [7]. We found two more practical problems with the built-in heading, in a pocket and in a car (Section 3.2).

**Vehicle speed from vibration.** Road vibration changes with speed, and learned models can read speed from accelerometer data alone [8]. Our speed model is of that kind, but it learns continuously from the phone’s own trips and must work for a phone in a trouser pocket on a motorcycle as well as in a car.

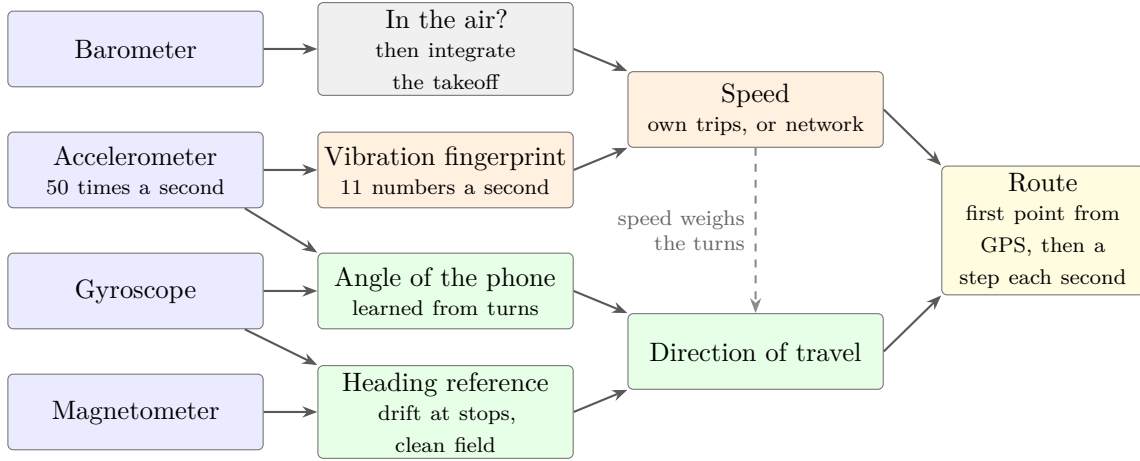
## 3 Method

A route is a series of short steps, one per second. Each step needs a length and a direction; with GPS both come from satellites, here both come from the phone (Figure 1).

### 3.1 Speed

**A fingerprint of the vibration.** Every second the phone takes the last five seconds of up-and-down acceleration, measured 50 times a second, and reduces it to eleven numbers: the energy of the shaking in nine frequency ranges and two measures of its overall strength. A vehicle shakes differently at different speeds, and the fingerprint captures that.

**A store of the phone’s own trips.** Whenever GPS is available the phone stores fingerprints with the speed GPS measured at that moment, keeping up to 4,000. Without GPS it looks



**Figure 1.** The system. Speed comes from the accelerometer’s vibration, or in the air from integrating the takeoff; direction combines the angle of the phone, learned from the gyroscope’s turns and the push felt in them, with a heading reference. GPS gives the first point and nothing else.

up the 12 stored fingerprints closest to the current one and averages their speeds, nearer ones counting more:

$$\hat{v} = \frac{\sum_{j=1}^{12} v_j / d_j^2}{\sum_{j=1}^{12} 1 / d_j^2}, \quad (1)$$

where  $v_j$  is a stored speed and  $d_j$  the distance between fingerprints. If even the closest stored fingerprint is far away, the phone gives no answer rather than a guess. Averaging pulls every answer toward the middle, so a correction learned from the stored examples stretches the answers back out.

**A starting network.** A new phone has no examples, so until it holds 3,000 it uses a small network (11 inputs, two layers of 32, one output) trained on recorded trips and shipped with the app. It needs nothing from the current trip, and it declines a fingerprint unlike anything it was trained on.

**Stops and hands.** A stopped vehicle is recognised from the absence of shaking, not from the speed model, which would read an idling engine as a crawl. A hand on the phone shakes it, and shaking reads as speed, so while the phone is being handled the speed may fall but not rise, and a phone held in the hand keeps the last speed measured before it was picked up.

**In the air.** A smooth cabin reads as standing still. In the air the phone instead measures the takeoff: from the last moment it was still, it follows its own turning with the gyroscope and adds up the push the accelerometer feels for up to two minutes, then holds the speed it reached. Core Motion’s own orientation cannot be used for this, because it takes a takeoff’s steady push for a tilt [7].

### 3.2 Direction

**A heading reference.** The phone’s built-in heading follows a direction inside the phone that depends on how it is held, and when that direction points nearly straight down — a phone head-down at an angle in a trouser pocket — it scattered by about  $49^\circ$  a minute although the phone did not move. The system therefore computes its own heading from the phone’s

orientation, using whichever edge of the phone lies closest to level; in the same position that heading scattered by about  $1^\circ$  a minute.

**Drift, measured at stops.** In a car the heading also drifts slowly with the gyroscope's own bias, because the steel around it keeps the magnetometer from holding north: on one drive it turned  $42^\circ$  in 25 minutes with the phone fixed in place. A stopped car does not turn, so any turning the heading shows while the car is stopped and the phone still is drift. The phone measures it as a rate, at most  $3^\circ$  a minute, and takes the accumulated drift off every heading.

**The magnetometer, when the field can be trusted.** Core Motion also sets north once at the start, often wrongly inside a vehicle, and then follows the gyroscope. On one drive it began  $20\text{--}26^\circ$  off while the magnetometer, read directly, put the phone within  $1\text{--}5^\circ$  of the truth. So whenever the magnetic field looks like Earth's — between 25 and  $65\ \mu\text{T}$ , with its strength and dip steady for a minute — the phone compares the magnetometer's heading with its own and takes the difference, less the drift already measured, as the correction to where the heading started. Beside a MagSafe charger the phone read up to  $2,600\ \mu\text{T}$  and nothing is taken from it.

**The angle of the phone, from turns.** The heading says where the phone points, not where the vehicle goes; the two differ by an angle  $\beta$  that depends on how the phone sits. In a turn the vehicle is pushed toward the inside of the curve, with a push equal to speed times turn rate, so comparing the sideways push the phone feels with the turning the gyroscope measures shows which way is forward. Written as complex numbers,

$$\hat{\beta} = \arg \left( \frac{\sum_R z_t}{\sum_R |v_t \omega_t|} + \frac{\sum_L z_t}{\sum_L |v_t \omega_t|} \right), \quad z_t = \frac{a_t}{|a_t|} i v_t \omega_t, \quad (2)$$

where  $a_t$  is the horizontal acceleration relative to the heading, written as its sideways and forward parts,  $v_t$  the speed,  $\omega_t$  the turn rate, and  $R$  and  $L$  the seconds turning right and left. Braking into a corner and pulling away out of it lean the answer one way in right turns and the other way in left ones, so the two kinds are averaged apart. Only seconds above 14 km/h count, and only when the turn explains at least a tenth of the push; otherwise the push is braking or throttle, and a reversing car's push points backwards. The first estimate comes after 30 seconds of turning, and whenever the estimate improves the part of the ride already drawn is redrawn with it.

**Walking.** Over four seconds the back-and-forth of the steps lines up with the direction of walking, and each push-off is sharper than each landing, which tells which end of that line is forward.

### 3.3 The route

After the first point, each second's step is laid from the previous point along the current direction, for the distance the speed covers in that second. No later GPS position, course or speed enters the route, the heading or the decisions between walking and riding; the magnetic declination is taken once, at the first point. Appendix B lists every setting.

**Table 1.** The recordings. GPS km is the distance GPS could check; half-minutes are the stretches used for speed; direction is graded in recordings with at least 50 graded seconds.

|            | Record-ings | Hours | GPS km | Half-minutes | Direction rec. (s) | Routes | Phone                 |
|------------|-------------|-------|--------|--------------|--------------------|--------|-----------------------|
| Motorcycle | 39          | 11.1  | 177    | 818          | 40 (13,248)        | 41     | trouser pocket        |
| Car        | 35          | 8.6   | 163    | 733          | 25 (8,996)         | 27     | pocket, flat or mount |
| Walking    | 17          | —     | 1.9    | 45           | 7 (721)            | —      | pocket                |
| Plane      | 1           | 0.24  | 74     | —            | 1 (876)            | —      | cabin                 |

## 4 Data and evaluation

### 4.1 Recordings

All recordings were made by the author with one iPhone, mostly in one city, with the phone’s raw sensor readings saved — acceleration, rotation, orientation and, in the most recent ones, the magnetic field, 50 times a second — and GPS recorded alongside. Table 1 summarises them. A motorcycle and a car were told apart from the sensors themselves: in a turn a motorcycle leans and the phone leans with it, while a car stays level and the phone is pushed sideways; this agreed with every journey whose vehicle was known. Walking is the time the app was counting steps.

### 4.2 Replaying the newest version

So that every result describes the newest version, each recording was replayed through the current method from its raw sensor data:

- **Speed:** the same fingerprint, the same 12-neighbour average, refusals and correction, and the same rules for stops, a phone in the hand, and holding the last answer. The store holds 4,000 examples drawn at random from every *other* journey, so no journey is measured with its own GPS in the store. On twelve recent journeys the replay came out typically 4% shorter than what the app itself had drawn.
- **Direction:** the heading reference, drift correction, magnetometer correction (where the magnetic field was recorded), the angle learned from turns and the redrawing, with the replayed speed.
- **Starting network:** each recording measured by a network trained without it (the recordings split into eight groups), and the store tested holding only 60 to 4,000 examples.
- **Walking:** the step detector replayed on the recorded acceleration with the stride the phone had learned.

Three inputs are taken from the recordings as they were: whether each second was walking or riding, Apple’s motion classifier, and the car-park ramp flag. Cabin pressure was not recorded, so on the flight the moment the app would call it airborne was taken from the same integration’s climb rate.

### 4.3 Measures

In a pocket GPS is itself accurate only to 10–40 m, so it is used over stretches long enough for that not to matter.

- **Speed and distance** are compared over half-minute stretches between GPS fixes better

**Table 2.** Speed by GPS speed band: number of half-minutes, the middle GPS and app speeds, and the average error (km/h).

| GPS band (km/h) | Motorcycle |      |      |       | Car      |      |      |       |
|-----------------|------------|------|------|-------|----------|------|------|-------|
|                 | <i>n</i>   | GPS  | App  | Error | <i>n</i> | GPS  | App  | Error |
| 0–10            | 253        | 2.6  | 2.5  | 4.6   | 229      | 2.6  | 2.9  | 4.2   |
| 10–20           | 159        | 14.7 | 19.2 | 9.0   | 125      | 15.4 | 16.7 | 7.0   |
| 20–30           | 138        | 25.0 | 26.8 | 7.4   | 114      | 25.2 | 25.1 | 6.2   |
| 30–40           | 109        | 35.1 | 30.9 | 7.2   | 97       | 35.1 | 34.1 | 6.5   |
| 40–50           | 79         | 44.0 | 35.2 | 12.5  | 58       | 43.7 | 45.4 | 9.7   |
| 50–60           | 43         | 53.2 | 38.4 | 17.2  | 42       | 54.6 | 55.0 | 9.8   |
| 60–80           | 34         | 66.4 | 41.2 | 29.4  | 56       | 68.9 | 61.8 | 11.8  |
| 80+             | 3          | 98.4 | 41.3 | 70.1  | 12       | 87.1 | 70.7 | 46.9  |
| All             | 818        | 19.9 | 21.2 | 9.0   | 733      | 21.1 | 22.9 | 7.3   |

than 40 m: the straight distance GPS moved against the distance the system added. For walking, only straight stretches between fixes better than 25 m are used.

- **Direction** is compared every second: the bearing from the GPS position 15 s earlier to the one 15 s later, against the system’s heading over the same half-minute (10 s for the plane).
- **Whole routes** are compared by finding the rotation and the size change that best lay the drawn route over the GPS track, pinned at the shared start. The rotation is how far the route is turned; the size change how much too big or small it is; the drift is how far the drawn end lies from GPS’s, as a share of the distance travelled.

#### 4.4 Stretches left out

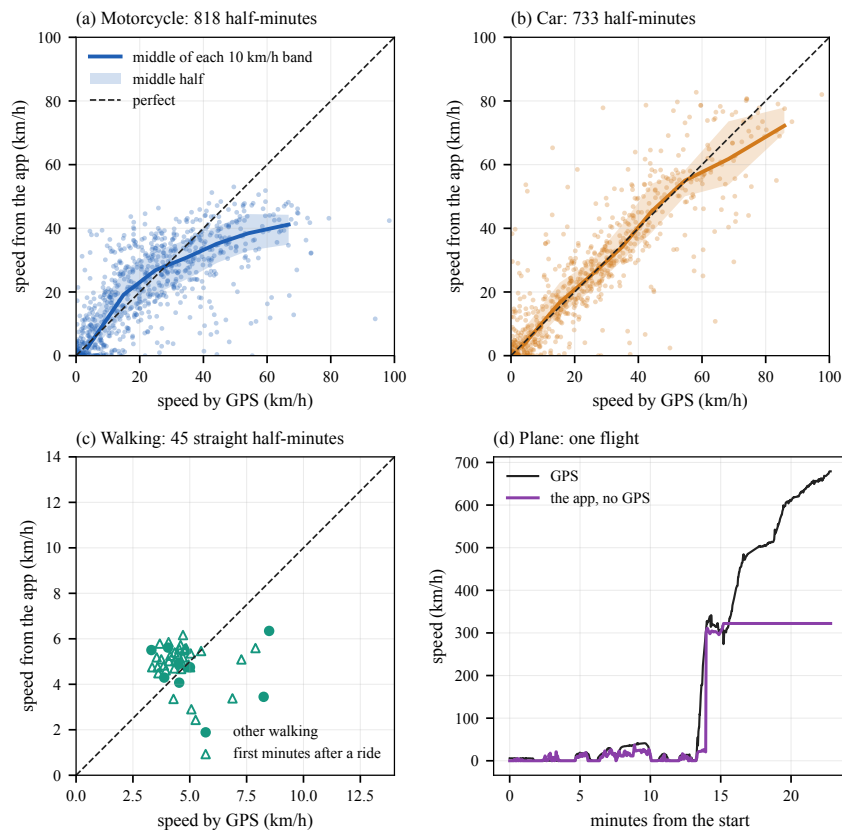
Four car journeys went into, started in or ended in underground car parks. There GPS kept claiming 10 m accuracy for positions it did not have — frozen while parked, wandering on the ramps — so those stretches, found from the barometer, are left out of every comparison. So are half-minutes that mix walking and riding, and those in which the app counted steps while GPS showed riding speed.

## 5 Results

### 5.1 Speed

Figure 2 and Table 2 compare speed half-minute by half-minute.

- **Motorcycle** (818 half-minutes). The app squeezes the range: 10–20 km/h reads about 19, 20–40 km/h is close, and at 60–80 km/h it reads about 41 where GPS measured 66. The squeeze comes from the averaging: when a fingerprint could belong to several speeds, the average lands in the middle. The average error is 9.0 km/h.
- **Car** (733 half-minutes). Much closer: from 10 to 60 km/h the middle reading is within about 2 km/h of GPS. Above that it reads low (62 where GPS measured 69, and 71 where it measured 87). The average error is 7.3 km/h.
- **Walking** (45 straight half-minutes). About 5.0 km/h where GPS measured 4.6.
- **Plane.** While taxiing the app reads about 17 km/h where GPS measured 18. In the air it measures the takeoff — 265 km/h forty seconds into the roll, where GPS measured 266



**Figure 2.** Speed from the app against GPS. (a, b) Each dot is a half-minute of riding; the line is the middle of each 10 km/h band and the shading the middle half. On the dashed line the two agree. (c) Straight half-minutes of walking; triangles are the first minutes after a ride. (d) One flight.

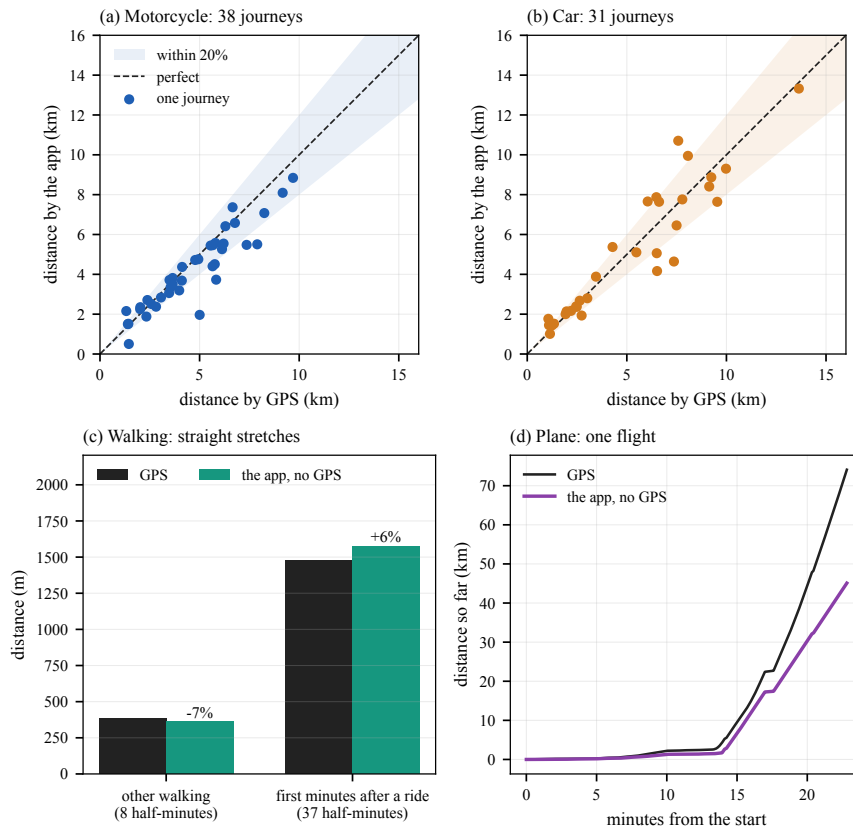
**Table 3.** Distance against GPS. Total is the whole category; the median journey, the share within 10% and 20%, and the range use journeys with at least 1 km checked.

|            | Journeys | GPS<br>km | App<br>km | Total  | Median<br>journey           | Within<br>10% | Within<br>20% | Range<br>app/GPS |
|------------|----------|-----------|-----------|--------|-----------------------------|---------------|---------------|------------------|
| Motorcycle | 39       | 176.5     | 158.4     | −10.3% | −4.7%                       | 45%           | 79%           | 0.35–1.63        |
| Car        | 35       | 163.1     | 162.7     | −0.3%  | −1.6%                       | 42%           | 65%           | 0.63–1.67        |
| Walking    | 45 str.  | 1.87      | 1.93      | +3.7%  | after a ride                | +6.5%         | otherwise     | −6.8%            |
| Plane      | 1        | 74        | 45        | −39.1% | takeoff measured, then held |               |               |                  |

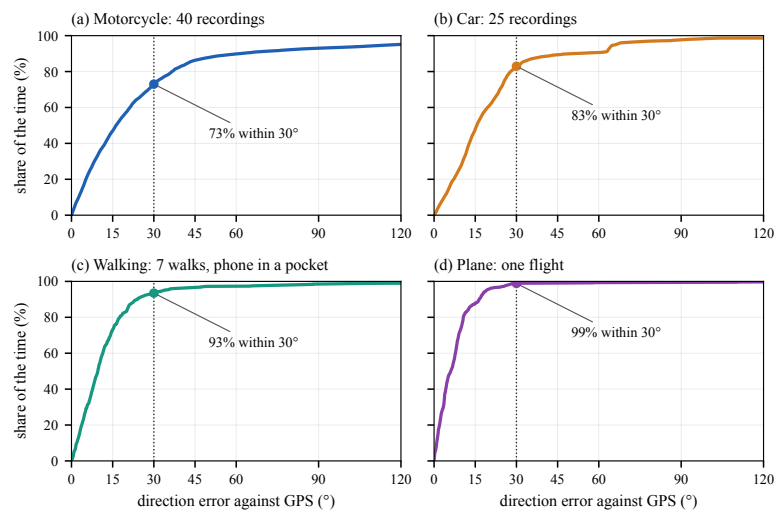
— and holds the 317 km/h it reached; it does not see the aircraft speed up to 680 km/h in the climb.

## 5.2 Distance

On a motorcycle the app added 158 km where GPS measured 177 km, 10% short: the distance lost at high speed outweighs what is gained at low speed. In a car it added 163 km of 163 km. Journey by journey the spread is wider (Table 3): 79% of motorcycle journeys and 65% of car journeys came within 20% of GPS. Walking came out 4% long, 7% long in the first minutes after a ride and 7% short at other times. The flight counted 45 of 74 km. Motorcycle and car together, the app added 321 km where GPS measured 340 km, 5.5% short.



**Figure 3.** Distance by the app against GPS. (a, b) Each dot is one journey with at least 1 km that GPS could check; the shading is within 20%. (c) Straight stretches of walking, added up. (d) Distance so far on the flight.



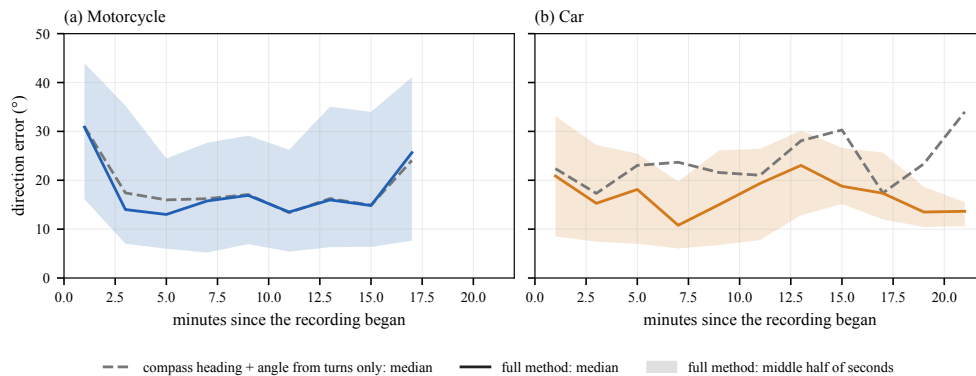
**Figure 4.** How often the direction is within a given angle of GPS, compared every second with a half-minute of GPS track (ten seconds for the plane).

### 5.3 Direction

Direction holds up in every case (Figure 4, Table 4). The median error is  $16^\circ$  on a motorcycle and in a car,  $10^\circ$  on foot and  $6^\circ$  in the air; 73%, 83%, 93% and 99% of seconds are within  $30^\circ$ . The tails differ: one motorcycle second in ten is off by more than  $61^\circ$ , against  $51^\circ$  in a car,

**Table 4.** Direction error against GPS: the median, the 75th and 90th percentiles, and the share of graded seconds within 15°, 30° and 45°.

|            | Graded            | Median | 75th pct. | 90th pct. | Within 15° | Within 30° | Within 45° |
|------------|-------------------|--------|-----------|-----------|------------|------------|------------|
| Motorcycle | 40 rec., 13,248 s | 16°    | 32°       | 61°       | 47%        | 73%        | 86%        |
| Car        | 25 rec., 8,996 s  | 16°    | 26°       | 51°       | 48%        | 83%        | 89%        |
| Walking    | 7 walks, 721 s    | 10°    | 16°       | 24°       | 73%        | 93%        | 97%        |
| Plane      | 1 flight, 876 s   | 6°     | 10°       | 16°       | 88%        | 99%        | 99%        |

**Figure 5.** Direction error through a ride, in two-minute bins with at least 300 graded seconds: the full method against the compass heading with the angle learned from turns alone.

because a phone in a trouser pocket moves as the rider shifts.

Figure 5 follows the error through a ride. In the first minutes the angle of the phone is still being learned and the error is highest; it settles within a few minutes. In a car the full method stays below the compass-and-turns method for most of the drive, as drift correction and the magnetometer keep the heading reference in place; on a motorcycle the two are close, since a motorcycle’s stops rarely hold the phone still enough to measure drift.

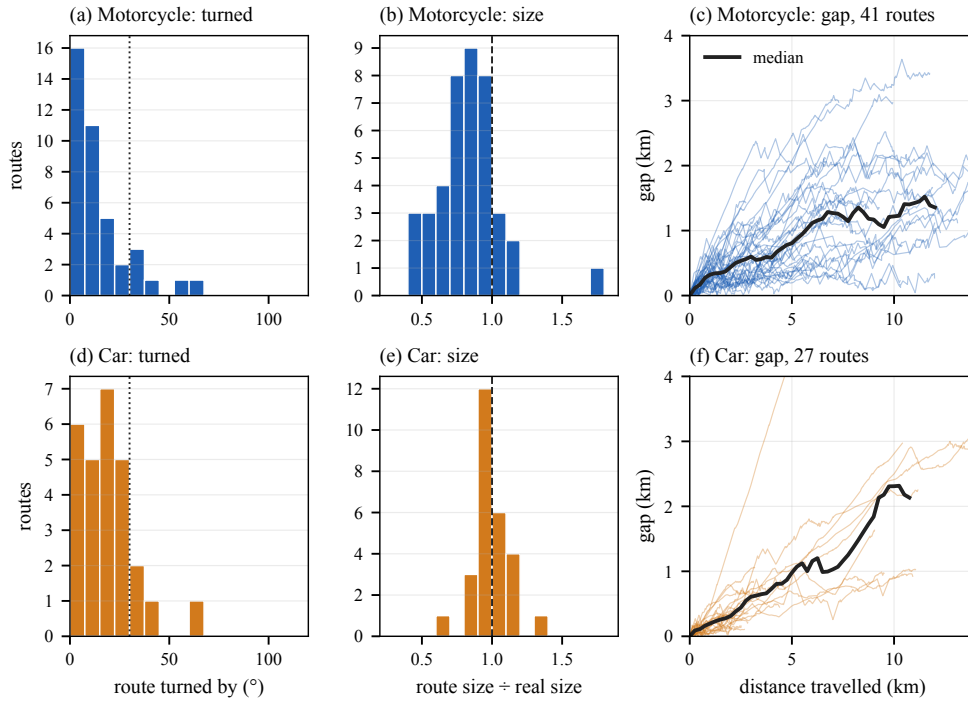
## 5.4 Whole routes

Figure 6 and Table 5 put speed and direction together. The median motorcycle route is turned 9° and 83% are within 30°; the median car route is turned 16° and 85% are within 30°. Motorcycle routes come out small — a median 0.84 of the real size — partly because distance is short and partly because small direction errors fold the path; car routes have the right size (median 0.98). The drawn end lies a median 17% (motorcycle) and 19% (car) of the distance travelled from GPS’s. Four routes are turned more than 45°; three of them are under 3 km, with too few turns to learn the angle of the phone well.

Figure 7 shows eight routes, chosen across the range of accuracy rather than the best ones: for each vehicle the routes at the 10th, 40th, 70th and 95th percentile of end drift.

## 5.5 What each part of the direction method adds

Table 6 replays every recording with the direction method built up one part at a time, graded exactly as above. For cars, drift measured at stops and the averaging of right and left turns matter most: together they take the share of seconds within 30° from 64% to 80% and routes within 30° from 70% to 85%, and the magnetometer adds three more points. For motorcycles every version is within a point or two. The check that a turn explains the push changes little on the road but decides the flight: without it the aircraft’s taxiing brakes set a wrong angle and



**Figure 6.** Routes drawn by the app against GPS. (a, d) How far each whole route is turned; the dotted line is  $30^\circ$ . (b, e) Its size relative to GPS. (c, f) How far the drawn position is from GPS as the journey goes on: one thin line per route, and the median.

**Table 5.** Whole routes against GPS.

|            |        |        | Whole route turned |                 |                 |              | Size   |            | End drift |
|------------|--------|--------|--------------------|-----------------|-----------------|--------------|--------|------------|-----------|
|            | Routes | GPS km | median             | $\leq 15^\circ$ | $\leq 30^\circ$ | $> 45^\circ$ | median | within 20% | median    |
| Motorcycle | 41     | 351    | $9^\circ$          | 66%             | 83%             | 3            | 0.84   | 54%        | 17%       |
| Car        | 27     | 146    | $16^\circ$         | 41%             | 85%             | 1            | 0.98   | 93%        | 19%       |

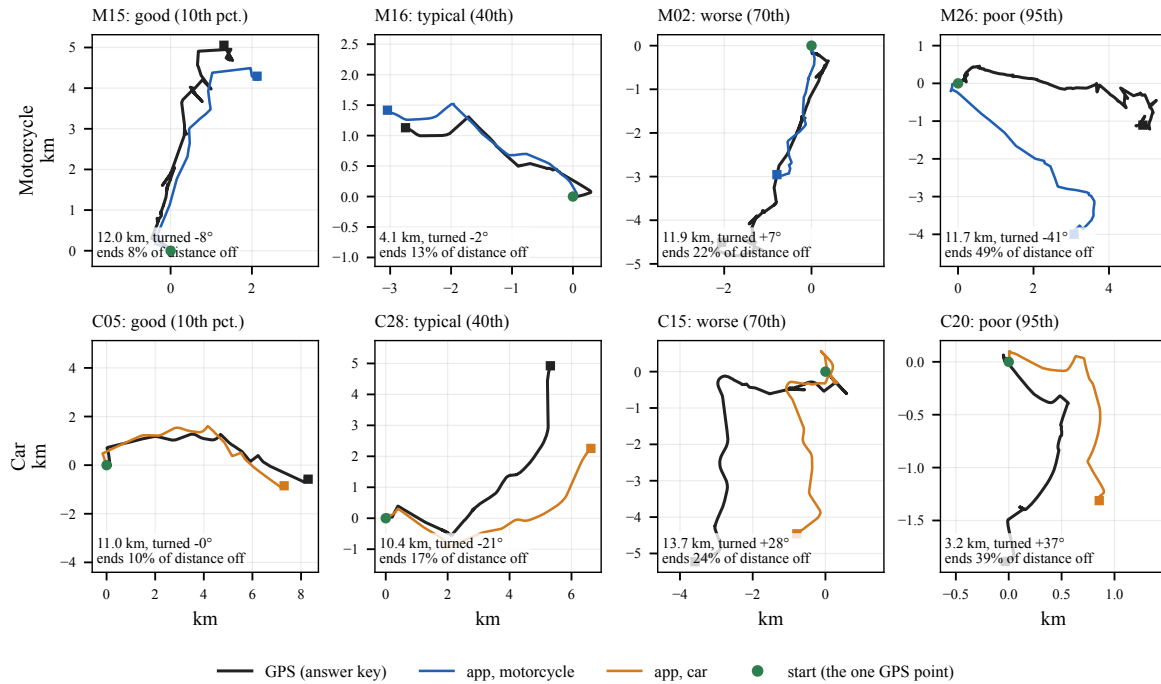
only 25–36% of the flight’s seconds are within  $30^\circ$ ; with it, 99%.

## 5.6 The flight

Figure 8 follows the flight with the same direction method as on the road. While taxiing the median error was  $9^\circ$  and in the air  $4^\circ$ . Ten seconds were more than  $30^\circ$  off: the pushback, when the aircraft moves tail first while the method gives the way the nose points, and nine while it was stopped or crawling before the runway, where the GPS track has no clear direction. In the climb the error stays small, about  $1^\circ$  by the end of the recording.

## 5.7 A new phone

A new phone starts with the built-in network. On recordings it had never seen it missed by 9.1 km/h on a motorcycle and 6.7 km/h in a car, against 9.0 and 7.3 for the main model with a full store, and counted 91.5% and 101.7% of the distance against 90.0% and 99.7% (Figure 9). A store still filling up does worse (Table 7): 10.6 km/h with 60 examples, 8.8 with 1,000, and level with the network at about 3,000 — which is where the app hands over to the store, which then keeps adapting to the phone’s own vehicle and pocket. Walking (103.7%) and the flight (61%) are the same with either, since they are counted by steps and by the measured takeoff.



**Figure 7.** Eight routes drawn without GPS (colour) against GPS (black), chosen at the 10th, 40th, 70th and 95th percentile of end drift for each vehicle. Labels match Appendix A. To avoid revealing where they were recorded, each pair is turned together by an arbitrary angle, which leaves the comparison unchanged.

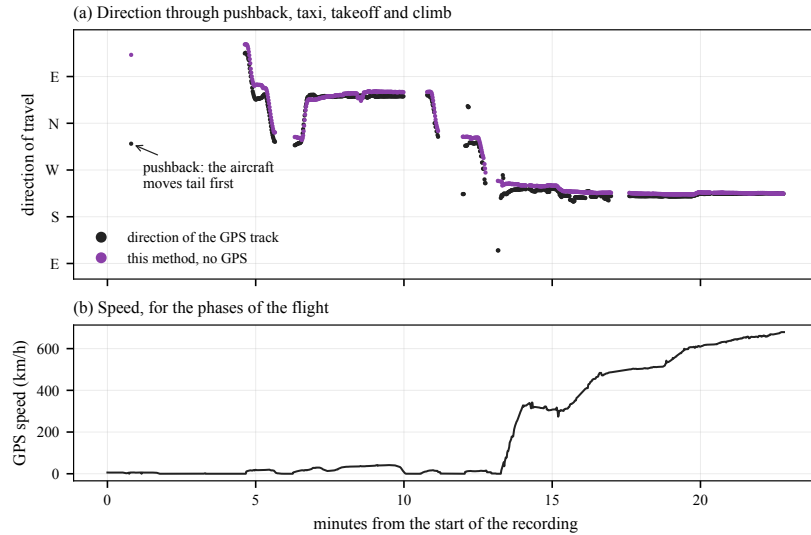
**Table 6.** What each part of the direction method adds: the share of graded seconds and of whole routes within  $30^\circ$ , routes turned more than  $45^\circ$ , and the flight's seconds within  $30^\circ$ . Each row adds one part to the row above. The magnetometer applies only where the magnetic field was recorded (one car drive so far).

| Direction method           | Motorcycle |        | Car     |        | Routes over $45^\circ$ | Flight seconds |
|----------------------------|------------|--------|---------|--------|------------------------|----------------|
|                            | Seconds    | Routes | Seconds | Routes |                        |                |
| Heading + angle from turns | 73%        | 85%    | 64%     | 70%    | 8                      | 25%            |
| + drift at stops           | 72%        | 85%    | 72%     | 78%    | 8                      | 28%            |
| + right, left apart        | 73%        | 85%    | 80%     | 85%    | 3                      | 36%            |
| + turn explains push       | 73%        | 83%    | 80%     | 85%    | 4                      | 99%            |
| + clean magnetometer       | 73%        | 83%    | 83%     | 85%    | 4                      | 99%            |

## 6 Discussion

### 6.1 Where it goes wrong

- **On a motorcycle, fast riding reads slow and slow riding reads fast.** Above 60 km/h the speed is under two-thirds of the true value (Table 2). Over a journey the two partly cancel, but a journey that is mostly fast comes out short.
- **The heading's starting point.** When the magnetic field cannot be trusted for the whole drive, an error Core Motion makes at the start stays for the drive: 24 of 65 recordings with enough GPS to grade were turned by more than  $15^\circ$  this way. Removing it would raise the share of seconds within  $30^\circ$  from 77% to 87% overall; only the magnetometer, when clean, can see it.
- **The first minute of a ride.** Until the angle of the phone is learned the direction comes



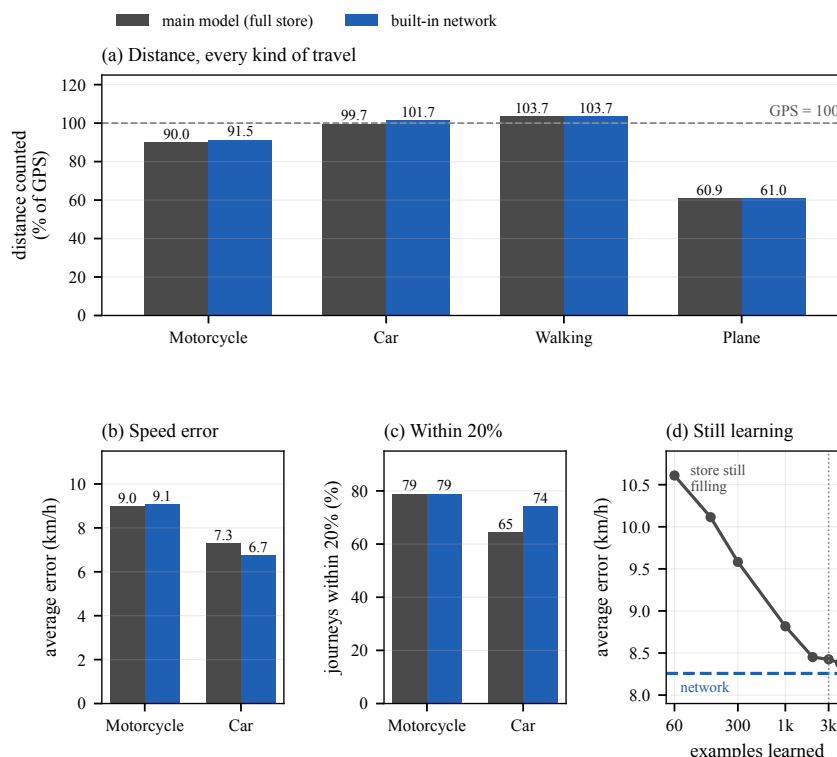
**Figure 8.** The flight, from the gate to the climb. (a) Direction of travel from the GPS track (black) and from this method, which uses no GPS (purple). (b) The aircraft’s speed from GPS.

**Table 7.** A store still filling up against the built-in network, over motorcycle and car journeys together, with each journey’s own examples left out.

| Speed model           | Average speed error (km/h) |            |     | Journeys |            |
|-----------------------|----------------------------|------------|-----|----------|------------|
|                       | All                        | Motorcycle | Car | Distance | within 20% |
| Store, 60 examples    | 10.6                       | 11.5       | 9.6 | −12.0%   | 51%        |
| Store, 150 examples   | 10.1                       | 11.1       | 8.9 | −17.3%   | 41%        |
| Store, 300 examples   | 9.6                        | 10.1       | 8.9 | −10.1%   | 49%        |
| Store, 1,000 examples | 8.8                        | 9.4        | 8.1 | −7.7%    | 57%        |
| Store, 2,000 examples | 8.5                        | 9.1        | 7.7 | −7.6%    | 65%        |
| Store, 3,000 examples | 8.4                        | 9.0        | 7.7 | −4.9%    | 68%        |
| Store, 4,000 examples | 8.4                        | 9.1        | 7.5 | −5.5%    | 69%        |
| Built-in network      | 8.3                        | 9.1        | 7.2 | −3.9%    | 75%        |

from the heading alone (Figure 5). The redrawing fixes the saved route afterwards, but a very short ride may never learn the angle well.

- **A phone that moves.** A hand shakes the phone and shaking reads as speed; a hand can no longer raise the speed, but a phone picked up while slowing keeps its last speed until it is still. A phone that shifts in the pocket turns the rest of the ride by about as much as it moved.
- **A magnet beside the phone.** By a car’s MagSafe charger the phone read up to 2,600  $\mu\text{T}$ , fifty times Earth’s field, yet reported its compass as well calibrated; the iPhone’s own Compass then drifted  $49^\circ$  with the gyroscope. The app ignores such a field, but it cannot then correct the heading.
- **Car-park ramps and rides never recognised.** A car crawling up a concrete ramp shakes like one at 40 km/h, so distance is withheld while climbing and turning together; once that test fired on an ordinary road and lost about 115 m. On one short ride Apple’s classifier never said “driving” and the step counter took the engine for footsteps: 59% short.
- **After the takeoff.** Holding the takeoff’s speed misses the speed-up in the climb: 317



**Figure 9.** The built-in network (blue) next to the main model with a full store (grey), on the same journeys, the network only on recordings it had never seen. (a) Share of the GPS distance counted. (b) Average speed error. (c) Journeys within 20%. (d) A store still filling up; the dotted line is where the app hands over.

held against 680 km/h reached, 61% of the distance. Examples learned on earlier flights with GPS would take over, which one flight could not test.

## 6.2 What did not work

Table 8 lists approaches that were built and measured against GPS on the same recordings and did no better. Two are worth a note. Leaving the speed out of the angle learner seemed attractive, since it would make direction independent of the speed model, but a car brakes into a corner and accelerates out of it, and a learner driven by the change in turn rate adds the two up instead of cancelling them. And weighting turns by the square root of their push drew individual drives very differently, better on one and worse on another, yet over 66 rides the learned angle was the same ( $9.5^\circ$  against  $9.2^\circ$  median error): single drives are a poor guide to a method.

## 6.3 Lessons

- **The phone’s own definitions caused more error than the models.** Its built-in heading becomes unreliable in a common pocket position, drifts in a car and can start wrong; its “user acceleration” points the opposite way to the vehicle’s acceleration. Each of these, once found, mattered more than any change to the estimators.
- **GPS can hide in decisions as well as in distance.** The no-GPS mode never drew a metre from GPS, yet at times it used GPS to decide between walking and riding, to set

**Table 8.** Approaches that were measured and did not do better.

| Approach                           | Idea                                              | Measured against GPS               |
|------------------------------------|---------------------------------------------------|------------------------------------|
| <i>Direction</i>                   |                                                   |                                    |
| Push when speeding up or braking   | In a straight line the push points forward        | 44° off                            |
| Axis the motorcycle leans about    | The lean axis follows the direction of travel     | 25° off                            |
| Direction of the rider's thigh     | The thigh points forward while seated             | 73° off                            |
| Angle from turns without the speed | Weight each second by the change in turn rate     | 15–40° off on one drive            |
| Angle from full physics            | Add the speed model's change in speed to the turn | cars 80% → 48% within 30°          |
| Square-root turn weights           | Let the many small bends count more               | a tie over 66 rides                |
| Median instead of mean for drift   | Ignore short bursts at stops                      | walking 93% → 88%                  |
| Looser stillness for drift         | Measure drift at more stops                       | motorcycles 73% → 68%              |
| Car magnetism fitted as a circle   | Remove the car's own field from the magnetometer  | no stable fit                      |
| <i>Walking, riding and the air</i> |                                                   |                                    |
| Thigh swing for walking direction  | Swing is sharper forwards than backwards          | wrong more often than right        |
| Other riding-or-walking tests      | A steady push; speed overruling steps             | blocked 5% of real walking         |
| Speed from turns in the air        | Sideways push over turn rate                      | phone's turning hid the aircraft's |
| Integrating the whole climb        | Carry the takeoff integration on                  | worked for one setting only        |
| <i>Speed</i>                       |                                                   |                                    |
| Last ten seconds into the network  | More context for each answer                      | better per second, not per journey |

the first heading, or to carry an angle learned before the mode was switched on. Only reading every place the mode touched GPS found them.

- **Two errors can hide each other.** On one drive a worse angle estimate drew a better route, because it cancelled an error in the heading reference. Judge a part of the method against physics, not only by the route it draws.
- **A replay must do everything the app does,** stops and a phone in the hand included; with those rules in, it comes within about 4% of what the app drew.

## 7 Limitations

All recordings come from one person and one iPhone, mostly in one city, with a single flight; other people, phones and vehicles may behave differently, the starting network most of all. Most results replay the newest method on earlier recordings rather than coming from live use. GPS in a pocket is accurate only to tens of metres, so small errors cannot be measured reliably. The magnetometer correction was recorded on one drive so far, beside a magnet for part of it.

## 8 Conclusion

A phone’s own sensors can draw a usable route without satellites. Direction is the stronger half: learned from the vehicle’s turns on a heading kept honest by its stops and, when possible, by the magnetometer, it stays within  $30^\circ$  of GPS most of the time on every kind of travel measured. Speed from vibration is close in a car and on foot but compressed on a motorcycle, and in the air the takeoff alone carries most of a flight. The largest remaining error is where the heading starts, which only a clean magnetic field can fix.

**Code and data.** The app, including the method described here, is open source at <https://github.com/yu314-coder/GPS-location-app>. The recordings contain the author’s locations and are not published.

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## A Every journey

Each motorcycle (M) and car (C) journey, in the order recorded, with no dates. Minutes and GPS km are what GPS could check; app/GPS is the distance ratio for journeys of at least 1 km; speed error is the average over half-minutes; direction is graded where at least 50 seconds could be; the route columns are the whole-route comparison. A dash means too little GPS to grade.

|     | Min. | GPS km | App/GPS<br>(%) | Speed err.<br>(km/h) | Dir. within<br>30° (%) | Dir. median<br>(°) | Route<br>km | Turned<br>(°) | Size |
|-----|------|--------|----------------|----------------------|------------------------|--------------------|-------------|---------------|------|
| M01 | 9    | 2.0    | 112            | 3.8                  | –                      | –                  | –           | –             | –    |
| M02 | 20   | 4.0    | 80             | 8.0                  | 64                     | 12                 | 11.9        | +7            | 0.56 |
| M03 | 14   | 4.8    | 99             | 6.0                  | 79                     | 23                 | 7.3         | +17           | 0.87 |
| M04 | 23   | 6.7    | 111            | 9.2                  | 65                     | 23                 | 11.9        | +23           | 0.99 |
| M05 | 18   | 5.7    | 96             | 10.1                 | 82                     | 12                 | 9.5         | –9            | 0.84 |
| M06 | 18   | 8.2    | 86             | 11.8                 | 89                     | 6                  | 12.0        | –3            | 1.00 |
| M07 | 16   | 5.8    | 96             | 7.9                  | 91                     | 19                 | 8.0         | –21           | 0.89 |
| M08 | –    | –      | –              | –                    | –                      | –                  | 2.5         | –57           | 0.58 |

|     | Min. | GPS km | App/GPS<br>(%) | Speed err.<br>(km/h) | Dir. within<br>30° (%) | Dir. median<br>(°) | Route<br>km | Turned<br>(°) | Size |
|-----|------|--------|----------------|----------------------|------------------------|--------------------|-------------|---------------|------|
| M09 | 4    | 1.4    | 107            | 5.3                  | 99                     | 3                  | 2.0         | -4            | 0.93 |
| M10 | 14   | 6.1    | 86             | 6.5                  | 36                     | 35                 | 7.3         | +33           | 0.85 |
| M11 | 19   | 6.2    | 89             | 8.7                  | 91                     | 7                  | 9.3         | -12           | 0.68 |
| M12 | 15   | 3.6    | 105            | 7.8                  | 77                     | 16                 | 8.8         | +7            | 0.74 |
| M13 | 10   | 2.4    | 114            | 7.3                  | 86                     | 7                  | 5.7         | +5            | 0.87 |
| M14 | 11   | 2.3    | 81             | 7.6                  | 79                     | 9                  | 6.5         | -1            | 0.75 |
| M15 | 20   | 3.5    | 88             | 9.8                  | 84                     | 8                  | 12.0        | -8            | 0.93 |
| M16 | 9    | 3.5    | 106            | 6.5                  | 97                     | 4                  | 4.1         | -2            | 1.16 |
| M17 | 27   | 9.7    | 91             | 9.3                  | 76                     | 10                 | 16.6        | -2            | 0.86 |
| M18 | —    | —      | —              | —                    | —                      | —                  | 2.7         | -113          | 0.47 |
| M19 | 14   | 2.0    | 116            | 7.5                  | 62                     | 19                 | 8.0         | -2            | 1.76 |
| M20 | 8    | 2.8    | 84             | 10.7                 | 92                     | 7                  | 4.0         | -9            | 0.73 |
| M21 | 9    | 0.3    | —              | 1.2                  | 25                     | 54                 | 3.2         | -30           | 0.47 |
| M22 | 25   | 7.9    | 70             | 10.0                 | 65                     | 16                 | 15.6        | -13           | 0.77 |
| M23 | 11   | 3.6    | 96             | 7.9                  | 77                     | 21                 | 4.3         | -19           | 0.86 |
| M24 | —    | —      | —              | —                    | 81                     | 17                 | —           | —             | —    |
| M25 | 15   | 1.4    | 107            | 7.0                  | 96                     | 9                  | 5.1         | -4            | 0.52 |
| M26 | 32   | 6.8    | 97             | 5.4                  | 41                     | 31                 | 11.7        | -41           | 0.94 |
| M27 | 12   | 2.6    | 98             | 5.8                  | 63                     | 17                 | 6.0         | -10           | 1.16 |
| M28 | 19   | 6.3    | 102            | 11.2                 | 86                     | 11                 | 13.9        | +10           | 0.81 |
| M29 | 21   | 4.1    | 90             | 5.7                  | 76                     | 16                 | 11.5        | -22           | 0.70 |
| M30 | 8    | 3.1    | 93             | 8.3                  | 84                     | 23                 | 5.7         | -11           | 1.03 |
| M31 | —    | —      | —              | —                    | 24                     | 52                 | 1.7         | +61           | 1.02 |
| M32 | 27   | 5.7    | 78             | 10.1                 | 58                     | 20                 | 13.1        | -3            | 0.83 |
| M33 | 20   | 4.9    | 96             | 12.3                 | 88                     | 12                 | 10.4        | -8            | 1.01 |
| M34 | 20   | 5.6    | 98             | 9.8                  | 80                     | 8                  | 10.3        | -2            | 1.00 |
| M35 | 20   | 5.8    | 64             | 13.3                 | 83                     | 12                 | 13.0        | -3            | 0.62 |
| M36 | 12   | 5.0    | 39             | 24.0                 | 30                     | 41                 | 6.6         | +36           | 0.46 |
| M37 | 20   | 3.5    | 95             | 8.0                  | 35                     | 38                 | 9.9         | -25           | 0.68 |
| M38 | 17   | 1.4    | 35             | 6.6                  | 64                     | 12                 | 5.1         | +7            | 0.74 |
| M39 | 16   | 1.3    | 163            | 7.3                  | 80                     | 19                 | 4.8         | -9            | 0.69 |
| M40 | 19   | 4.1    | 106            | 8.4                  | 52                     | 30                 | 12.3        | +7            | 0.77 |
| M41 | 20   | 5.8    | 78             | 11.0                 | 84                     | 14                 | 8.5         | +2            | 0.72 |
| M42 | 24   | 7.4    | 74             | 9.9                  | 68                     | 24                 | 13.8        | -22           | 0.96 |
| M43 | 27   | 9.2    | 88             | 10.1                 | 89                     | 13                 | 14.7        | -13           | 0.92 |
| C01 | 10   | 2.5    | 97             | 2.4                  | 86                     | 8                  | 2.8         | -12           | 0.98 |
| C02 | 15   | 2.2    | 98             | 2.8                  | 96                     | 10                 | 2.8         | +10           | 0.99 |
| C03 | 11   | 0.0    | —              | 1.1                  | —                      | —                  | 0.2         | -38           | 0.91 |
| C04 | 13   | 2.0    | 108            | 3.1                  | 100                    | 5                  | 2.7         | -2            | 1.06 |
| C05 | 32   | 9.2    | 96             | 4.0                  | 98                     | 11                 | 11.0        | -0            | 0.89 |
| C06 | 34   | 10.0   | 93             | 3.0                  | 98                     | 15                 | 11.2        | -16           | 0.92 |
| C07 | 11   | 6.6    | 115            | 10.4                 | —                      | —                  | —           | —             | —    |
| C08 | 16   | 7.6    | 141            | 13.6                 | —                      | —                  | —           | —             | —    |
| C09 | 14   | 8.1    | 123            | 19.1                 | —                      | —                  | —           | —             | —    |
| C10 | 9    | 2.7    | 71             | 12.1                 | —                      | —                  | —           | —             | —    |
| C11 | 13   | 6.5    | 64             | 17.1                 | —                      | —                  | —           | —             | —    |
| C12 | 24   | 7.5    | 86             | 9.4                  | —                      | —                  | —           | —             | —    |
| C13 | 16   | 6.5    | 78             | 12.1                 | —                      | —                  | —           | —             | —    |
| C14 | 29   | 6.0    | 127            | 8.0                  | —                      | —                  | —           | —             | —    |
| C15 | 25   | 9.5    | 80             | 8.4                  | 64                     | 19                 | 13.7        | +28           | 0.64 |
| C16 | 24   | 7.4    | 63             | 9.7                  | 88                     | 16                 | 10.9        | +17           | 0.97 |
| C17 | 3    | 0.8    | —              | 3.1                  | 100                    | 5                  | 1.6         | +6            | 1.11 |
| C18 | 21   | 6.5    | 121            | 7.4                  | 97                     | 6                  | 9.0         | -6            | 1.16 |
| C19 | 12   | 3.5    | 113            | 4.2                  | 98                     | 5                  | 4.4         | -6            | 1.15 |
| C20 | 10   | 2.5    | 95             | 3.5                  | 69                     | 25                 | 3.2         | +37           | 0.85 |

|     | Min. | GPS km | App/GPS (%) | Speed err. (km/h) | Dir. within 30° (%) | Dir. median (°) | Route km | Turned (°) | Size |
|-----|------|--------|-------------|-------------------|---------------------|-----------------|----------|------------|------|
| C21 | 12   | 2.6    | 102         | 3.4               | 71                  | 26              | 3.1      | -26        | 0.90 |
| C22 | 10   | 1.3    | 113         | 7.0               | 49                  | 33              | 2.1      | -26        | 0.81 |
| C23 | 6    | 1.1    | 89          | 10.7              | 57                  | 12              | 1.6      | +16        | 0.93 |
| C24 | 7    | 0.9    | –           | 3.6               | –                   | –               | 2.0      | +21        | 1.03 |
| C25 | 8    | 1.1    | 132         | 4.6               | 76                  | 25              | 2.7      | +36        | 0.91 |
| C26 | 19   | 7.8    | 100         | 6.5               | 95                  | 15              | 9.5      | -10        | 0.98 |
| C27 | 6    | 1.2    | 116         | 5.3               | 79                  | 16              | 2.5      | +8         | 1.00 |
| C28 | 15   | 9.1    | 92          | 6.0               | 97                  | 23              | 10.4     | -21        | 1.00 |
| C29 | 25   | 13.6   | 98          | 5.8               | 84                  | 26              | 16.3     | -26        | 0.96 |
| C30 | 8    | 1.1    | 167         | 9.4               | 96                  | 11              | 1.8      | +20        | 1.09 |
| C31 | 4    | 0.9    | –           | 8.4               | 100                 | 7               | 1.2      | +7         | 1.34 |
| C32 | 15   | 3.0    | 93          | 5.3               | 56                  | 29              | 4.7      | +29        | 1.02 |
| C33 | 12   | 4.3    | 125         | 7.0               | 99                  | 8               | 4.9      | +10        | 1.20 |
| C34 | 11   | 1.9    | 104         | 2.8               | 98                  | 17              | 2.9      | -16        | 1.01 |
| C35 | 18   | 5.5    | 93          | 3.1               | 1                   | 64              | 6.5      | +64        | 0.93 |

## B Settings

**Table 10.** Every setting of the method as evaluated.

| Setting                              | Value                                                                   |
|--------------------------------------|-------------------------------------------------------------------------|
| <i>Speed</i>                         |                                                                         |
| Vibration window for one fingerprint | 5 s at 50 Hz, 11 numbers                                                |
| Store of the phone’s own examples    | up to 4,000                                                             |
| Neighbours averaged                  | 12                                                                      |
| Built-in network                     | 11 → 32 → 32 → 1; used until the store holds 3,000                      |
| Network refuses a fingerprint        | farther from all training clusters than 99.5% of training data          |
| Takeoff integration in the air       | from the last still moment, up to 120 s, then held                      |
| <i>Direction</i>                     |                                                                         |
| Riding begins                        | 9 km/h (2.5 m/s)                                                        |
| Learning the phone’s angle           | above 14 km/h (4 m/s); turn explains $\geq 10\%$ of the push            |
| First estimate                       | after 30 s and a summed speed $\times$ turn rate of 20 m/s <sup>2</sup> |
| Redrawing a ride                     | when the angle moves $\geq 3^\circ$ , at most every 10 s                |
| A new ride                           | after 20 s without riding                                               |
| A phone taken out of the pocket      | angle forgotten after 10 s in the hand                                  |
| Drift measured at stops              | stopped, turning $< 0.005$ rad/s, hand-rotation $< 0.2$ rad/s           |
| Drift estimate                       | after 30 still seconds, 10-minute memory, at most $3^\circ/\text{min}$  |
| Clean magnetic field                 | 25–65 $\mu\text{T}$ , strength and dip steady to 3 over 60 s            |
| Magnetometer correction              | 60 clean seconds in the last 20 min, spread $< 15^\circ$                |
| Walking direction window             | 4 s                                                                     |